Experiment #3
Active Filter and Oscillators

In this experiment you will design, build and test a Wien-bridge oscillator and an active low pass filter. In your designs you will use the 741 op-amp. Your plans should include complete designs to satisfy the specifications given below. Take into account any real properties of the 741 op-amp which significantly affect your designs. As part of your design, your plans should include complete analyses of the transfer functions for your filters (with derivations) and pole/zero diagrams. The frequency response of the filter designed in this lab can be calculated using any mathematical software package. You could use MatLab's function *FREQS* to provide an especially simple way of plotting the amplitude and phase response of any filter. Although you will predict both the magnitude of phase response for the filters, you will only measure the magnitude response.

Deliverables:

1. Pre-Lab 3A (Part A)
2. Pre-Lab 3B (Part B)
3. Lab Report

Part A : Low Pass Filters

Prelab 3A:
In the prelab part, try to come up with a transfer function of the LPF. Use Matlab to plot amplitude and phase response of the transfer function.

Using the circuit in Figure 1, design, build and test two low pass filters:

Filter 1	**Filter 2**
$C_1 = C_2$	**$R_B = 0$, $R_A = \infty$**
	(e.g., the op-amp is a voltage follower)
f_o = 380Hz	**f_o = 380Hz**
Q = .707	**Q = 1**

Please note: For both filters choose $R_1 = R_2$. In your tests measure magnitude frequency responses and confirm these with your theoretical expectations. Are the gains of the filters what they should be?

For Filter 1 only, measure the effect of changing R_B (e.g., gain of the op-amp) on the magnitude frequency response by re-evaluating magnitude frequency response for $R_B \cong 15\%$ above and below the original value you chose. How do variations of gain affect the Q and magnitude frequency response?

For Filter 2 what problems do you face in component selection if you want a higher Q in the range of 10 or more?

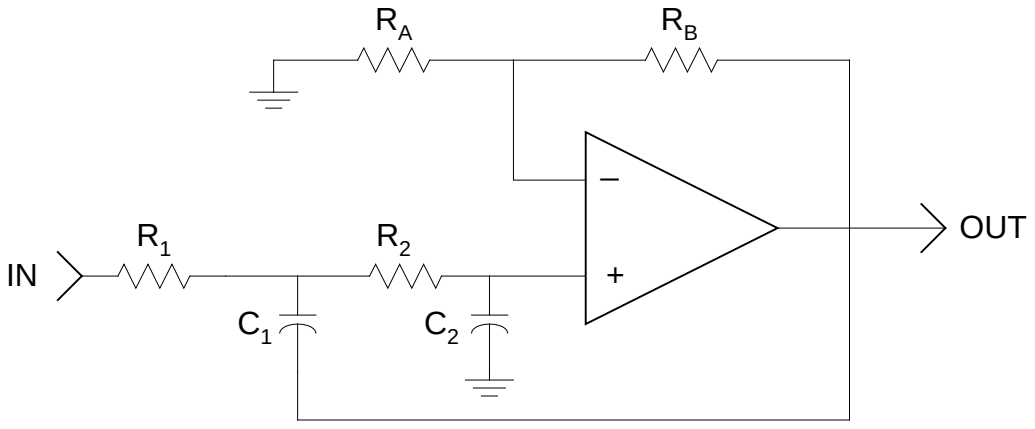

Figure 1. Low Pass Active Filter

Part B : Oscillators

Apply the Barkhausen criterion to determine the condition and the frequency of oscillation for the Wien-bridge oscillator shown in Figure 3. Build and test the oscillator you have designed. Note that R_1 or R_2 should be composed of a fixed resistor plus potentiometer, as you will want to fine adjust its value (and therefore the value of R_2/R_1) to produce oscillations. By changing R_2/R_1, you see the oscillations vanish or grow. The amplitudes of these oscillations may readily exceed the maximum output voltage of the op-amp.
Vary your potentiometer and set it to the point where oscillations start to occur. Measure a) their frequency and b) measure the actual resistances that are necessary to barely produce oscillations. Compare the ratio of resistances and the frequency with your prior estimate based on the Barkhausen criterion.
For very large oscillations (you observe nearly square waves), did the frequency shift?

Prelab 3B:
For the prelab please calculate the frequency of oscillation and the minimum value of R_2/R_1 for oscillation to occur.

Build the stabilization circuit (described in the lab notes). Why does this circuit stabilize the oscillator?

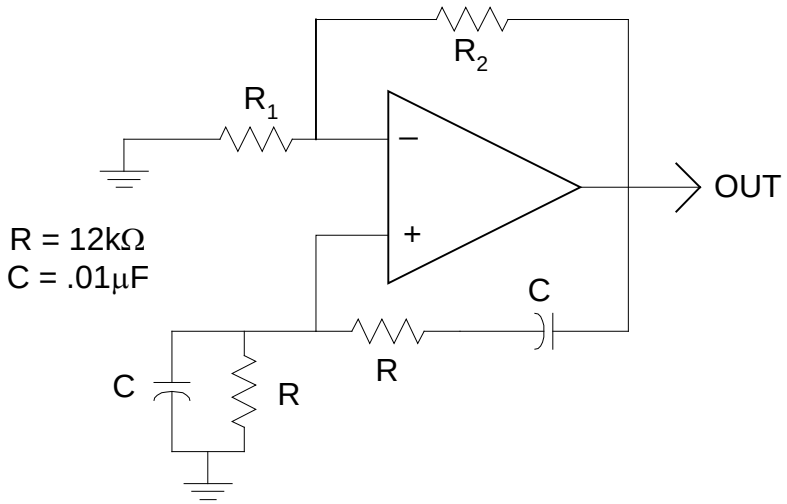

Figure 3. Wien-bridge Oscillator